

Where the Herglotz rational values meet the golden ladders and the Clausen lattice

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469ff48bb80ab3c4e221417ab9ac2934d8f3d22a. Companion to `herglotz-rational-values`; bridges to
`golden-polylog-ladders` and `clausen-roots-of-unity`. Reproducer
`src/PolyLog/tools/scratch/herglotz/bridges.py`.

Abstract

The general formula $F(2/q) - F(1) = \cdots - \frac{1}{2} \sum \text{Li}_2(-\cot^2 \frac{\pi k}{q})$ introduces dilogarithms at algebraic arguments; for $q = 5$ these live in $\mathbb{Q}(\sqrt{5})$, the field of our golden polylogarithm ladders, and the Herglotz *derivative* law introduces a Clausen sum $W(q)$ that lives in our conductor- q Clausen lattice. We make both connections explicit. The golden-argument dilogarithms of $F(2/5)$ split into a symmetric part that reduces to $\text{Li}_2(\frac{1}{5})$ plus golden constants and an antisymmetric part that is a genuinely new $\mathbb{Q}(\sqrt{5})$ real-dilogarithm constant; and the derivative Clausen sum $W(q)$ is an element of the conductor- q Clausen lattice over the real cyclotomic field $\mathbb{Q}(\zeta_q)^+$, with clean closed forms whenever $\cot \frac{2\pi k}{q} \in \mathbb{Q}(\zeta_q)^+$. All identities are verified to $\sim 10^{-80}$.

1 The golden bridge ($q = 5$)

For $q = 5$ the two algebraic arguments are $-\cot^2 \frac{\pi}{5} = -\frac{5+2\sqrt{5}}{5}$ and $-\cot^2 \frac{2\pi}{5} = -\frac{5-2\sqrt{5}}{5}$, both in $\mathbb{Q}(\sqrt{5})$. Comparing our $F(2/5)$ formula with Radchenko–Zagier’s eq. (24) (which uses the rational $\text{Li}_2(4/5)$) gives a dilogarithm functional equation tying the two:

Proposition 1. *With $\ell_k := \log^2(2 \sin \frac{\pi k}{5})$,*

$$\text{Li}_2\left(\frac{4}{5}\right) + \text{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \text{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = 2 \log^2 2 - 2 \log 2 \log \frac{2}{5} - 6(\ell_1 + \ell_2).$$

Eliminating the rational dilogarithm gives a closed form for the symmetric golden-argument sum:

Proposition 2. *With $\varphi = \frac{1+\sqrt{5}}{2}$,*

$$\text{Li}_2\left(-\cot^2 \frac{\pi}{5}\right) + \text{Li}_2\left(-\cot^2 \frac{2\pi}{5}\right) = \text{Li}_2\left(\frac{1}{5}\right) - \frac{\pi^2}{6} - 3 \log^2 \varphi + \frac{1}{4} \log^2 5.$$

Substituting Proposition ?? into the $F(2/5)$ theorem re-expresses that Herglotz value entirely through $\text{Li}_2(\frac{1}{5})$ and golden constants — the same constants the golden ladders reach for from the other side.

Remark 3 (A new $\mathbb{Q}(\sqrt{5})$ constant). Only the *symmetric* combination reduces. The antisymmetric part $X := \frac{1}{2}(\text{Li}_2(-\cot^2 \frac{\pi}{5}) - \text{Li}_2(-\cot^2 \frac{2\pi}{5}))$ admits no integer relation over $\{\pi^2, \log^2 \varphi, \log^2 5, \log^2 2, \text{Li}_2(\frac{1}{5}), \text{Li}_2(\frac{2}{5}), \log \text{ products}\}$ (PSLQ with a Champenowne canary, 100 digits, maxcoeff = 10^5): it is a genuinely new real-dilogarithm constant attached to $\mathbb{Q}(\sqrt{5})$. (Consistently, X is not balanced as a Bloch-group element, so Borel's vanishing of $K_3(\mathbb{Q}(\sqrt{5})) \otimes \mathbb{Q}$ does not force it to reduce.)

2 The Clausen bridge (the derivative law)

The derivative law $\frac{2}{q}F'(2/q) - (1 + \log 2) = \frac{q^2-4}{24q}\pi^2 + W(q)$ carries the Clausen sum $W(q) = \sum_{k=1}^{q-1} \cot \frac{2\pi k}{q} \text{Cl}_2(\frac{2\pi k}{q})$. Unlike the *value* $F(2/q)$, this does not reduce to logarithms: $W(q)$ is a $\mathbb{Q}(\zeta_q)^+$ -linear combination of the conductor- q Clausen values $\text{Cl}_2(\frac{2\pi k}{q})$ — i.e. an element of our conductor- q Clausen lattice over the real cyclotomic field. The coefficients $\cot \frac{2\pi k}{q}$ are rational over $\mathbb{Q}(\zeta_q)^+$ exactly when $\sin \frac{2\pi k}{q} \in \mathbb{Q}(\zeta_q)^+$; in those cases $W(q)$ has a clean closed form, e.g.

$$W(8) = 2 \text{Cl}_2\left(\frac{\pi}{4}\right) - 2 \text{Cl}_2\left(\frac{3\pi}{4}\right), \quad W(12) = \frac{22\sqrt{3}}{9} \left(\text{Cl}_2\left(\frac{\pi}{6}\right) - \text{Cl}_2\left(\frac{5\pi}{6}\right) \right).$$

For $q = 5, 7, 9, 11$ the cotangents lie in a strictly larger field (e.g. $\cot \frac{2\pi}{5} \notin \mathbb{Q}(\sqrt{5})$), so no rational- or $\mathbb{Q}(\sqrt{5})$ -coefficient reduction exists, while membership in the conductor- q Clausen lattice over $\mathbb{Q}(\zeta_q)^+$ persists. So differentiating F at a rational moves from the real-dilogarithm world of $F(2/q)$ into the imaginary-dilogarithm (Clausen) lattice we mapped at roots of unity.

Verification

Propositions ??-??, the irreducibility of X , and the $W(8), W(12)$ closed forms are checked to $\sim 10^{-80}$ in `src/PolyLog/tools/scratch/herglotz/bridges.py`.